I should just reiterate here that the arrow of time and the Second Law of Thermodynamics are so integral to the functioning of the universe that if there's no way for entropy to go up, *nothing can happen*. It is no longer possible for any organized structures to exist, for any evolution to happen, for any meaningful processes of any kind to occur. A necessary part of anything really happening is energy moving from one place to another. If entropy can't go up, then energy can't flow from one place to another without immediately flowing back, erasing anything that might have just, by chance, appeared to occur. Energy *gradients* are the basis of life, but also of any other structure or machine that performs any kind of work. Energy gradients can't exist in a universe that is just one giant (but very cold) heat bath. Heat is useless. Heat is death.

There are some caveats.

And to be clear, these are not caveats of the "well, technically there's this small detail" sort, but caveats of the "OMG this changes everything" sort.

This time, the weirdness comes down to a part of physics called *statistical mechanics*. This is what we use when we need to talk about something like temperature, which is really just the amount of motion in a system of particles, without painstakingly describing the path of each and every particle individually. Statistical mechanics is where the Second Law really shines, since it lets you describe a big complicated system in terms of one important property: its entropy. But it also introduces a kind of "out." That point about how it's an inescapable law of the universe that the entropy always increases? That technically only applies on average over sufficiently large scales. On the *quantum* scale, or even on large scales if you wait long enough, unpredictable fluctuations will, from time to time, spontaneously shift some part of the system into a lower-entropy state at random. The larger the system, the less likely it is that fluctuations could do much of anything at all, but in a universe that is in an eternal expansion and contains only a cosmological constant, there's a lot of time and space for waiting around, and even extremely low-probability events happen. It's *unlikely* that a whale and a bowl of petunias could suddenly pop into existence in completely empty space, but, in principle, if you wait long enough, it could happen.

This might come in handy. If anything can spontaneously pop into existence after the Heat Death, why not another universe?